

Lecture 7. Iterative Linear Quadratic Regulator

Discrete Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Linear Quadratic Regulator

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$x_{k+1} = A_k x_k + B_k u_k$$

$Q_0, \dots, Q_{N-1} \quad S_N$ symmetric positive semi-definite

$R_0, \dots, R_{N-1}$ symmetric positive definite

A_k, B_k controllable

Linear Quadratic Regulator

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$x_{k+1} = A_k x_k + B_k u_k$$

Hamiltonian

$$H_k(x_k, u_k, \lambda_{k+1}) = \frac{1}{2} x_k^T Q_k x_k + \frac{1}{2} u_k^T R_k u_k + \lambda_{k+1}^T (A_k x_k + B_k u_k)$$

Maximum principle

$$\lambda_k = \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1}) = Q_k x_k + A_k^T \lambda_{k+1}$$
$$0 = \frac{\partial}{\partial u} H_k(x_k, u_k, \lambda_{k+1}) = R_k u_k + B_k^T \lambda_{k+1}$$

Linear Quadratic Regulator

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$x_{k+1} = A_k x_k + B_k u_k$$

Boundary condition

$$J_N^*(x_N) = \frac{1}{2} x_N^T S_N x_N \quad \lambda_N = \frac{\partial}{\partial x} J_N^*(x_N) = S_N x_N \quad \gamma_N = -\frac{1}{2} x_N^T S_N x_N$$

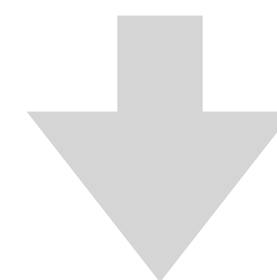
From N to N-1 step

$$\begin{aligned} \lambda_{N-1} &= Q_{N-1} x_{N-1} + A_{N-1}^T \lambda_N = Q_{N-1} x_{N-1} + A_{N-1}^T S_N A_{N-1} x_{N-1} + A_{N-1}^T S_N B_{N-1} u_{N-1} \\ 0 &= R_{N-1} u_{N-1} + B_{N-1}^T \lambda_N = R_{N-1} u_{N-1} + B_{N-1}^T S_N A_{N-1} x_{N-1} + B_{N-1}^T S_N B_{N-1} u_{N-1} \end{aligned}$$

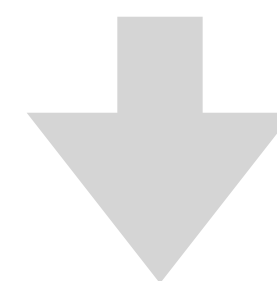
From N to N-1: Control Law

$$\lambda_{N-1} = Q_{N-1}x_{N-1} + A_{N-1}^T\lambda_N = Q_{N-1}x_{N-1} + A_{N-1}^TS_NA_{N-1}x_{N-1} + A_{N-1}^TS_NB_{N-1}u_{N-1}$$

$$0 = R_{N-1}u_{N-1} + B_{N-1}^T\lambda_N = R_{N-1}u_{N-1} + B_{N-1}^TS_NA_{N-1}x_{N-1} + B_{N-1}^TS_NB_{N-1}u_{N-1}$$



$$0 = [B_{N-1}^TS_NB_{N-1} + R_{N-1}]u_{N-1} + B_{N-1}^TS_NA_{N-1}x_{N-1}$$



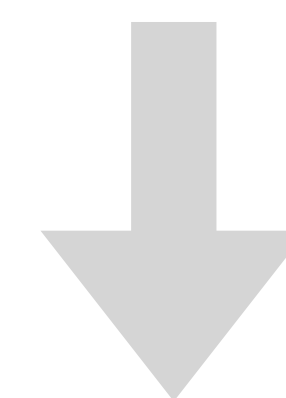
$$u_{N-1} = - \underbrace{[B_{N-1}^TS_NB_{N-1} + R_{N-1}]^{-1} B_{N-1}^TS_NA_{N-1}}_{K_{N-1}} x_{N-1}$$

*** We can take the inverse due to
positive definiteness assumptions**

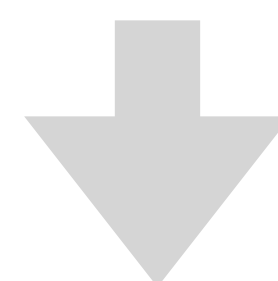
From N to N-1: Co-State

$$\lambda_{N-1} = Q_{N-1}x_{N-1} + A_{N-1}^T \lambda_N = Q_{N-1}x_{N-1} + A_{N-1}^T S_N A_{N-1}x_{N-1} + A_{N-1}^T S_N B_{N-1}u_{N-1}$$

$$u_{N-1} = - \underbrace{[B_{N-1}^T S_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T S_N A_{N-1}}_{K_{N-1}} x_{N-1}$$



$$\lambda_{N-1} = Q_{N-1}x_{N-1} + A_{N-1}^T S_N A_{N-1}x_{N-1} - A_{N-1}^T S_N B_{N-1}K_{N-1}x_{N-1}$$



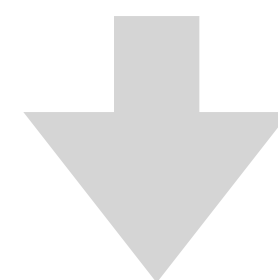
$$\lambda_{N-1} = \underbrace{[Q_{N-1} + A_{N-1}^T S_N A_{N-1} - A_{N-1}^T S_N B_{N-1}K_{N-1}]}_{P_{N-1}} x_{N-1}$$

$$P_N := S_N$$

$$P_{N-1} = Q_{N-1} + A_{N-1}^T P_N A_{N-1} - A_{N-1}^T P_N B_{N-1} [B_{N-1}^T P_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T P_N A_{N-1}$$

From N to N-1: Value Function

$$J_{N-1}^*(x_{N-1}) = \frac{1}{2}x_{N-1}^T Q_{N-1} x_{N-1} + \frac{1}{2}u_{N-1}^T R_{N-1} u_{N-1} + \frac{1}{2}x_N^T P_N x_N$$



$$u_{N-1} = -K_{N-1}x_{N-1}$$

$$x_N = A_{N-1}x_{N-1} + B_{N-1}u_{N-1} = [A_{N-1} - B_{N-1}K_{N-1}]x_{N-1}$$

$$\begin{aligned} J_{N-1}^*(x_{N-1}) = & \frac{1}{2}x_{N-1}^T \underline{Q_{N-1}} x_{N-1} + \frac{1}{2}x_{N-1}^T \underline{K_{N-1}^T R_{N-1} K_{N-1}} x_{N-1} \\ & + \frac{1}{2}x_{N-1}^T \underline{[A_{N-1} - B_{N-1}K_{N-1}]^T P_N [A_{N-1} - B_{N-1}K_{N-1}]} x_{N-1} \end{aligned}$$

We are going to show

$$P_{N-1} = Q_{N-1} + K_{N-1}^T R_{N-1} K_{N-1} + [A_{N-1} - B_{N-1}K_{N-1}]^T P_N [A_{N-1} - B_{N-1}K_{N-1}]$$

By definition

$$P_{N-1} = Q_{N-1} + A_{N-1}^T P_N A_{N-1} - A_{N-1}^T P_N B_{N-1} [B_{N-1}^T P_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T P_N A_{N-1}$$

From N to N-1: Value Function

Target Equation

$$\begin{aligned}
 P_{N-1} &= \cancel{Q_{N-1}} + \underbrace{K_{N-1}^T R_{N-1} K_{N-1}}_{\text{green}} + \underbrace{[A_{N-1} - B_{N-1} K_{N-1}]^T P_N [A_{N-1} - B_{N-1} K_{N-1}]}_{\text{red}} \\
 &\quad \parallel \\
 &\quad \underbrace{A_{N-1}^T \cancel{P_N} A_{N-1}}_{\text{orange}} - \underbrace{2A_{N-1}^T P_N B_{N-1} K_{N-1}}_{\text{yellow}} + \underbrace{K_{N-1}^T B_{N-1}^T P_N B_{N-1} K_{N-1}}_{\text{green}} \\
 &\quad \swarrow \quad \searrow \\
 &\quad \underbrace{K_{N-1}^T [R_{N-1} + B_{N-1}^T P_N B_{N-1}] K_{N-1}}_{\text{green}} \quad = \quad \underbrace{A_{N-1}^T P_N B_{N-1} K_{N-1}}_{\text{yellow}}
 \end{aligned}$$

By definition

$$P_{N-1} = \cancel{Q_{N-1}} + \underbrace{A_{N-1}^T \cancel{P_N} A_{N-1}}_{\text{orange}} - \underbrace{A_{N-1}^T P_N B_{N-1} [B_{N-1}^T P_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T P_N A_{N-1}}_{\text{red}}$$

K_{N-1}

From N to N-1: Value Function

- Hence we verified that

$$J_{N-1}^*(x_{N-1}) = \frac{1}{2} x_{N-1}^T P_{N-1} x_{N-1}$$

- Then we can conclude

$$\gamma_{N-1} = J_{N-1}^*(x_{N-1}) - \lambda_{N-1}^T x_{N-1} = -\frac{1}{2} x_{N-1}^T P_{N-1} x_{N-1}$$

- Co-state can only provide first-order information (i.e. gradient). To fully describe the value function, we still need to compute the zero-order term.

From N to N-1: Summary

$$u_{N-1} = - \underbrace{[B_{N-1}^T P_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T P_N A_{N-1}}_{K_{N-1}} x_{N-1}$$

$$P_{N-1} = Q_{N-1} + A_{N-1}^T P_N A_{N-1} - A_{N-1}^T P_N B_{N-1} [B_{N-1}^T P_N B_{N-1} + R_{N-1}]^{-1} B_{N-1}^T P_N A_{N-1}$$

$$\lambda_{N-1} = P_{N-1} x_{N-1}$$

$$\gamma_{N-1} = -\frac{1}{2} x_{N-1}^T P_{N-1} x_{N-1}$$

$$J_{N-1}^*(x_{N-1}) = \frac{1}{2} x_{N-1}^T P_{N-1} x_{N-1}$$

Induction

At time $k+1$ $J_{k+1}^*(x_{k+1}) = \frac{1}{2}x_{k+1}^T P_{k+1} x_{k+1}$ $\lambda_{k+1} = P_{k+1} x_{k+1}, \gamma_{k+1} = -\frac{1}{2}x_{k+1}^T P_{k+1} x_{k+1}$

Analogous to the case from N to $N-1$,

Solve the equations from the maximum principle

$$H_k(x_k, u_k, \lambda_{k+1}) = \frac{1}{2}x_k^T Q_k x_k + \frac{1}{2}u_k^T R_k u_k + \lambda_{k+1}^T (A_k x_k + B_k u_k)$$

$$\lambda_k = \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1}) = Q_k x_k + A_k^T \lambda_{k+1}$$

$$0 = \frac{\partial}{\partial u} H_k(x_k, u_k, \lambda_{k+1}) = R_k u_k + B_k^T \lambda_{k+1}$$

At time k

$$J_k^*(x_k) = \frac{1}{2}x_k^T P_k x_k \quad \lambda_k = P_k x_k, \gamma_k = -\frac{1}{2}x_k^T P_k x_k$$

$$P_k = Q_k + A_k^T P_{k+1} A_k - A_k^T P_{k+1} B_k [R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k$$

$$u_k = - \underbrace{[R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k}_{K_k} x_k$$

Discrete Algebraic Riccati Equation (DARE)

$$P_k = Q_k + A_k^T P_{k+1} A_k - A_k^T P_{k+1} B_k [R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k$$

- It describes the dynamics of the co-state
- Can be viewed as another form of Bellman update
- There is a unique solution of all P_k if we compute backward in time from the boundary condition $P_N = S_N$
- The solution does not depend on the system state, hence can be computed offline.

Implementation of LQR

- Obtain system dynamic model $x_{k+1} = A_k x_k + B_k u_k$
- Specify the matrices $Q_0, \dots, Q_{N-1}, S_N, R_0, \dots, R_{N-1}$ in the objective function

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

- Solve for $P_0, \dots, P_N$ in the DARE backward in time (offline)
- Real-time control: at every time step, once we obtain the state measurement x_k , then we apply the following optimal control

$$u_k = - \underbrace{[R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k}_{K_k} x_k$$

Example: Regulation of a Double Integrator

- Dynamics: one dimensional double integrator with sampling time Δt

$$x_{k+1} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix} u_k \quad \text{Q: What type of discretization is it?}$$

- Cost function

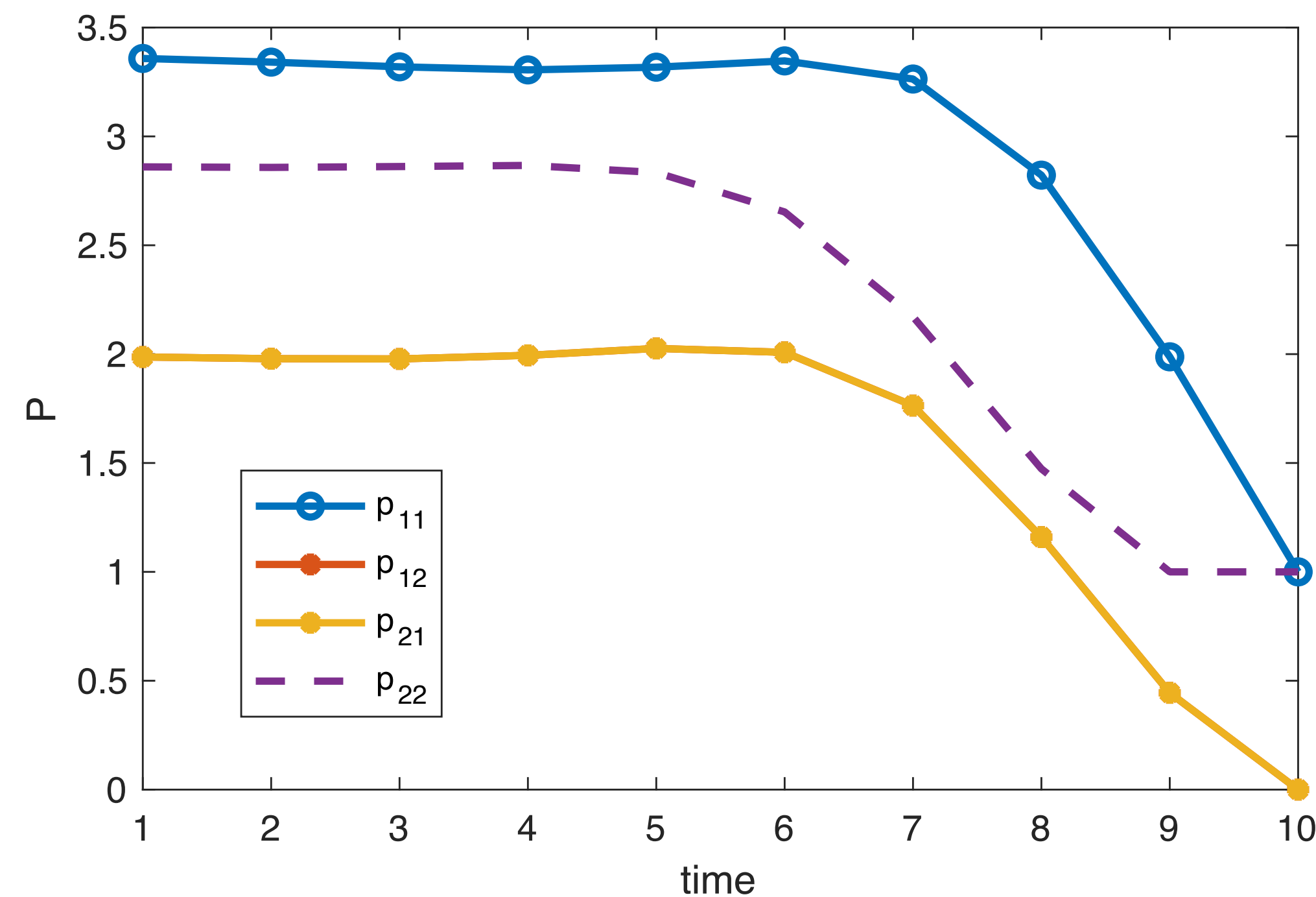
$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q x_k + u_k^T R u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, R = 1$$

Example: Regulation of a Double Integrator

- DARE

$$P_k = Q + A^T P_{k+1} A - A^T P_{k+1} B [R + B^T P_{k+1} B]^{-1} B^T P_{k+1} A$$

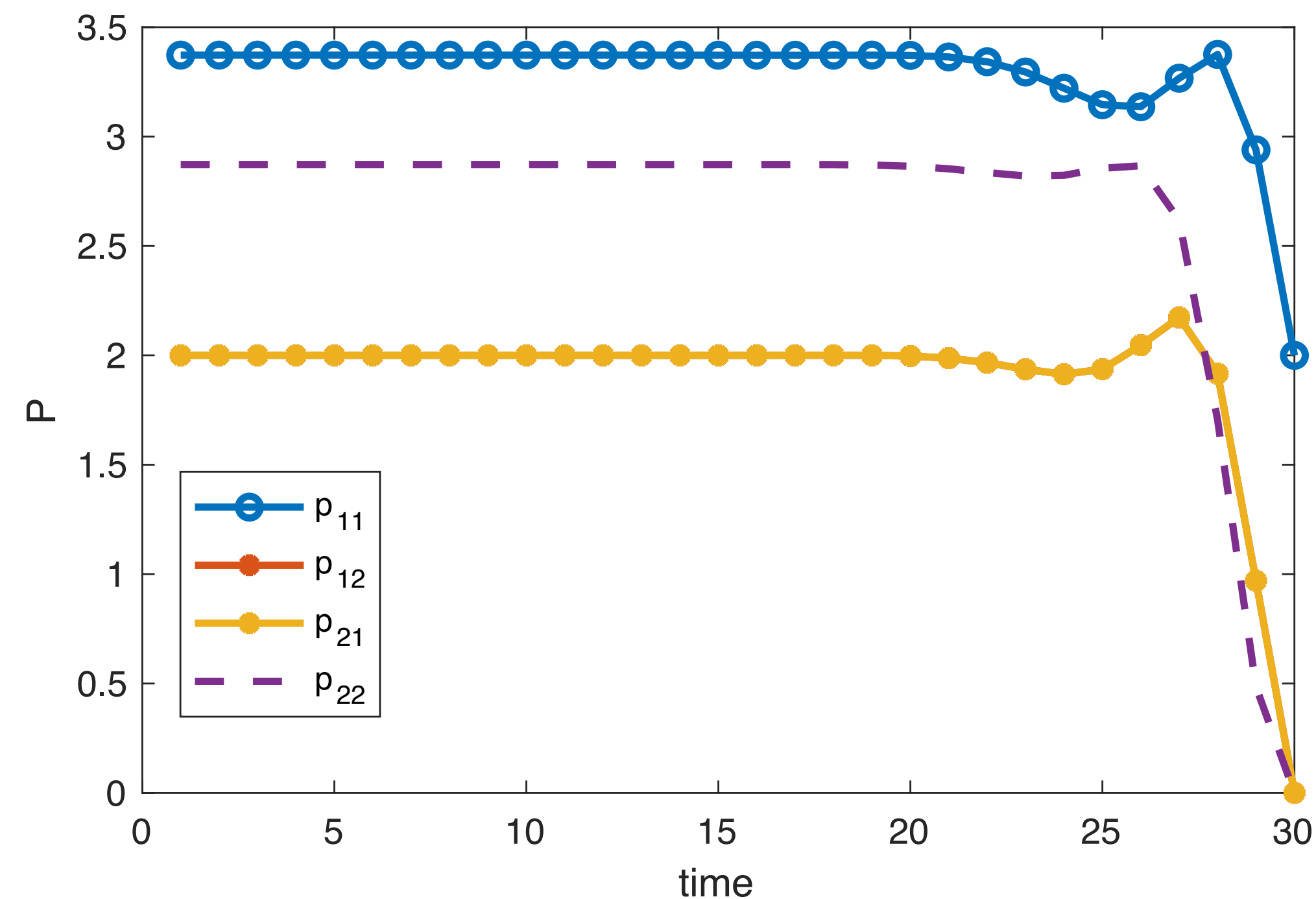


$$N = 10, P_N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example: Regulation of a Double Integrator

- DARE

$$P_k = Q + A^T P_{k+1} A - A^T P_{k+1} B [R + B^T P_{k+1} B]^{-1} B^T P_{k+1} A$$



$$N = 30, P_N = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Example: Regulation of a Double Integrator

- Observations
 - P_k is always symmetric
 - Regardless of the boundary condition P_N , the solution of the DARE always converges to the same steady state P_s
 - In this way, the control law converges (backward in time) to

$$u_k = - \underbrace{[R + B^T P_s B]^{-1} B^T P_s A}_{K_s} x_k$$

Example: Regulation of a Double Integrator

- The converged control law is indeed the solution to an **infinite horizon** LQR problem

$$u_k = - \underbrace{[R + B^T P_s B]^{-1} B^T P_s A}_{K_s} x_k$$

- There are two problems (to be answered later):
 - Under which condition the DARE (hence the control law) will converge?
 - Will the converged control law stabilize the system?

Continuous Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Linear Quadratic Regulator

$$J = \frac{1}{2} \int_{t=0}^T [x_t^T Q_t x_t + u_t^T R_t u_t] dt + \frac{1}{2} x_T^T S_T x_T$$

$$\dot{x} = A_t x + B_t u$$

Q_t, S_T symmetric positive semi-definite

R_t symmetric positive definite

A_t, B_t controllable

Linear Quadratic Regulator

$$J = \frac{1}{2} \int_{t=0}^T [x_t^T Q_t x_t + u_t^T R_t u_t] dt + \frac{1}{2} x_T^T S_T x_T$$

$$\dot{x} = A_t x + B_t u$$

Hamiltonian

$$H(t, x_t, u_t, \lambda_t) = \frac{1}{2} x_t^T Q_t x_t + \frac{1}{2} u_t^T R_t u_t + \lambda_t^T (A_t x_t + B_t u_t)$$

Maximum principle

$$\begin{aligned} -\dot{\lambda}_t &= \frac{\partial}{\partial x} H(t, x_t, u_t, \lambda_t) = Q_t x_t + A_t^T \lambda_t \\ 0 &= \frac{\partial}{\partial u} H(t, x_t, u_t, \lambda_t) = R_t u_t + B_t^T \lambda_t \quad \longrightarrow \quad u_t = -R_t^{-1} B_t^T \lambda_t \end{aligned}$$

Optimal Trajectory

- According to the maximum principle, the optimal trajectory satisfies the following ODE

$$\begin{aligned}\dot{x}_t &= A_t x_t - B_t R_t^{-1} B_t^T \lambda_t \\ \dot{\lambda}_t &= -Q_t x_t - A_t^T \lambda_t\end{aligned}$$

- with boundary condition x_0 and

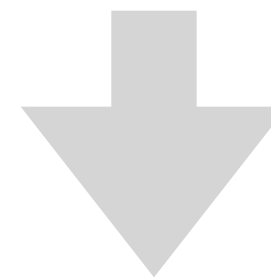
$$\lambda_T = \frac{\partial}{\partial x} l_T(x_T) = S_T x_T$$

Algebraic Riccati Equation

$$\begin{aligned}\dot{x}_t &= A_t x_t - B_t R_t^{-1} B_t^T \lambda_t \\ \dot{\lambda}_t &= -Q_t x_t - A_t^T \lambda_t\end{aligned}$$

- Assume that $\lambda_t = P_t x_t$ and $P_T = S_T$

$$\begin{aligned}0 &= \dot{\lambda}_t + Q_t x_t + A_t^T \lambda_t \\ &= \dot{P}_t x_t + \underline{P_t \dot{x}_t} + Q_t x_t + A_t^T \lambda_t \\ &= \dot{P}_t x_t + \underline{P_t A_t x_t} - \underline{P_t B_t R_t^{-1} B_t^T P_t x_t} + Q_t x_t + A_t^T P_t x_t\end{aligned}$$



$$\dot{P}_t + P_t A_t + A_t^T P_t - P_t B_t R_t^{-1} B_t^T P_t + Q_t = 0$$

Implementation of LQR

- Obtain system dynamic model $\dot{x} = A_t x + B_t u$
- Specify the matrices Q_t , S_T , and R_t in the objective function

$$J = \frac{1}{2} \int_{t=0}^T [x_t^T Q_t x_t + u_t^T R_t u_t] dt + \frac{1}{2} x_T^T S_T x_T$$

- Solve for P_t in the ARE backward in time (offline)
- Real-time control: at time t , we apply the following optimal control

$$u_t = -R_t^{-1} B_t^T P_t x_t$$

Comparison

Continuous Time

Discrete Time

System
Dynamics

$$\dot{x} = A_t x + B_t u$$

$$x_{k+1} = A_k x_k + B_k u_k$$

Cost Function

$$J = \frac{1}{2} \int_{t=0}^T [x_t^T Q_t x_t + u_t^T R_t u_t] dt + \frac{1}{2} x_T^T S_T x_T$$

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

Optimal
Control Law

$$u_t = -R_t^{-1} B_t^T P_t x_t$$

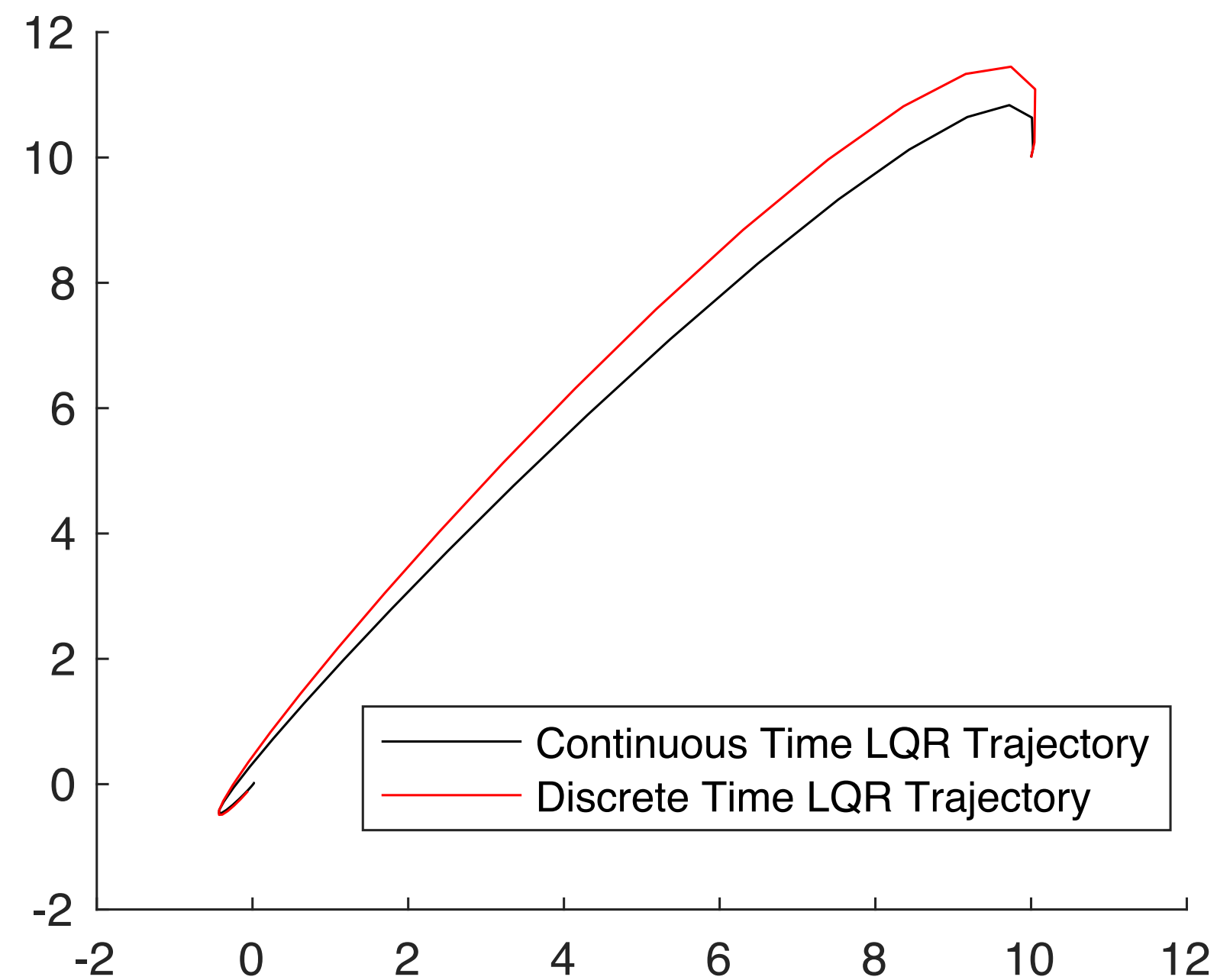
$$u_k = - \underbrace{[R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k}_{K_k} x_k$$

Riccati
Equation

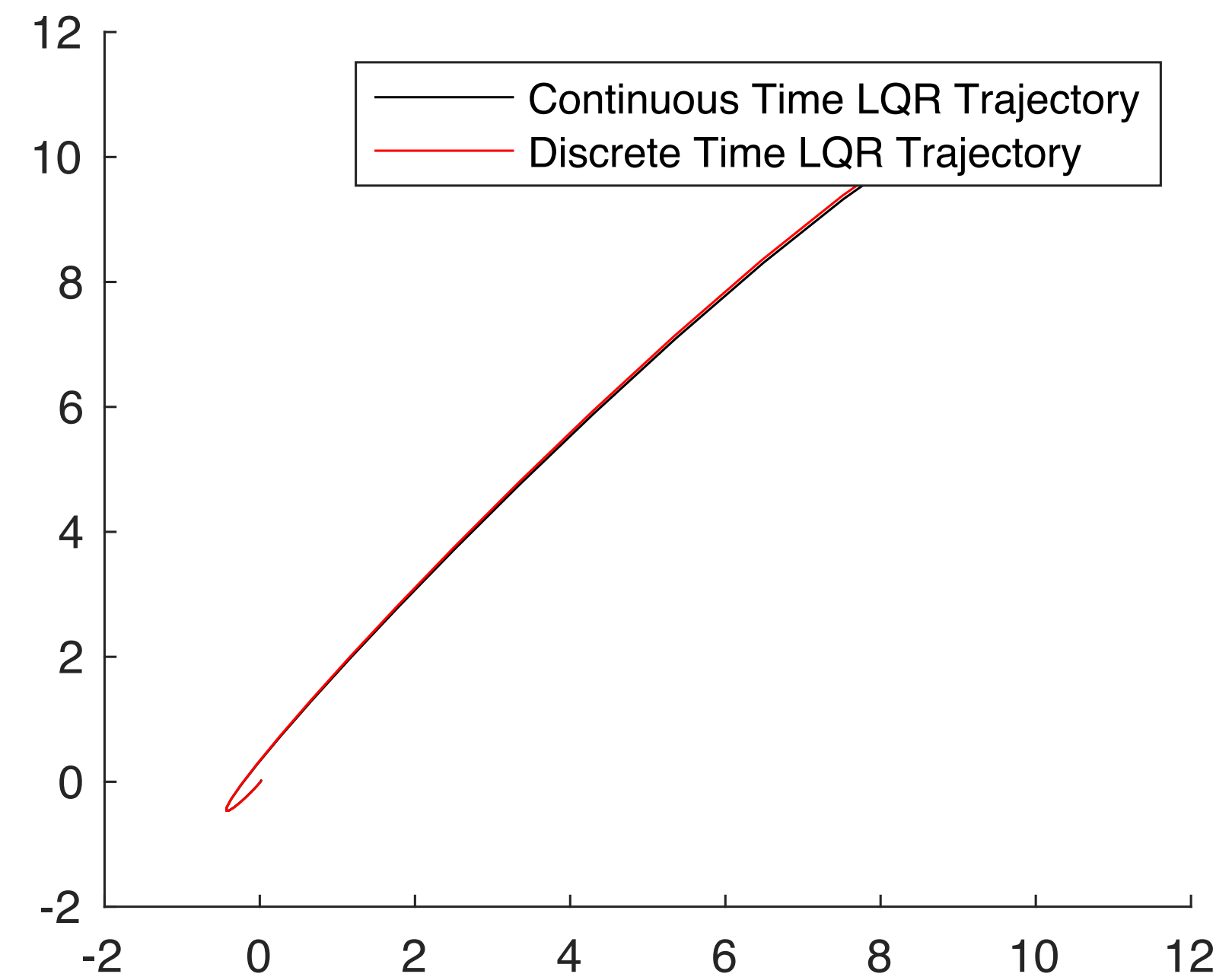
$$\dot{P}_t + P_t A_t + A_t^T P_t - P_t B_t R_t^{-1} B_t^T P_t + Q_t = 0$$

$$P_k = Q_k + A_k^T P_{k+1} A_k - A_k^T P_{k+1} B_k [R_k + B_k^T P_{k+1} B_k]^{-1} B_k^T P_{k+1} A_k$$

Comparison



Sample time = 1



Sample time = 0.1

Iterative Linear Quadratic Regulator

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

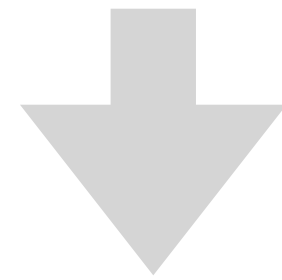
$$x_{k+1} = f(x_k, u_k)$$

- General idea:
 - Linearize the dynamics at reference trajectory and get the Δ dynamics
 - Solve the corresponding (time-varying) LQR problem with respect to the Δ dynamics to obtain the optimal Δ control sequence
 - Forward simulate the optimal Δ control sequence and get the new reference trajectory
 - Repeat the process until the reference trajectories converge

Linearize Dynamics

$$x_{k+1} = f(x_k, u_k) \quad \text{Reference} \quad (\bar{x}_k, \bar{u}_k)$$

$$\delta x_k := x_k - \bar{x}_k, \quad \delta u_k := u_k - \bar{u}_k$$



$$\delta x_{k+1} \approx A_k \delta x_k + B_k \delta u_k + d_k$$

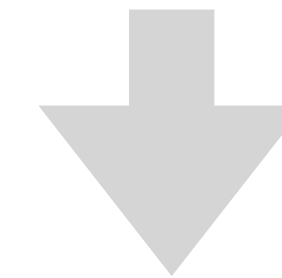
$$A_k = \left. \frac{\partial f}{\partial x} \right|_{(\bar{x}_k, \bar{u}_k)}, \quad B_k = \left. \frac{\partial f}{\partial u} \right|_{(\bar{x}_k, \bar{u}_k)}, \quad d_k = \underline{f(\bar{x}_k, \bar{u}_k) - \bar{x}_{k+1}}$$

Error in the reference trajectory

Example: Unicycle 3-State

$$x_k = [p_k^x; p_k^y; \theta_k] \quad u_k = [v_k; w_k]$$

$$x_{k+1} = x_k + \begin{bmatrix} \cos \theta_k \Delta t & 0 \\ \sin \theta_k \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} u_k$$



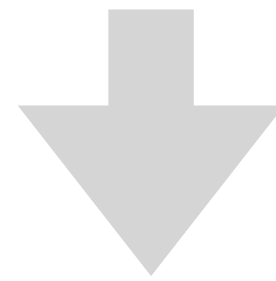
$$A_k = \begin{bmatrix} 1 & 0 & -\Delta t \bar{v}_k \sin \bar{\theta}_k \\ 0 & 1 & \Delta t \bar{v}_k \cos \bar{\theta}_k \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_k = \begin{bmatrix} \Delta t \cos \bar{\theta}_k & 0 \\ \Delta t \sin \bar{\theta}_k & 0 \\ 0 & \Delta t \end{bmatrix}$$

Δ Cost

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$\delta x_k := x_k - \bar{x}_k, \quad \delta u_k := u_k - \bar{u}_k$$



$$J = J_{\text{nominal}} + J_{\Delta}$$

$$J_{\Delta} = \frac{1}{2} \sum_{k=0}^{N-1} \left(\delta x_k^{\top} Q \delta x_k + \delta u_k^{\top} R \delta u_k + 2 \underbrace{\bar{x}_k^{\top} Q}_{q_k^{\top}} \delta x_k + 2 \underbrace{\bar{u}_k^{\top} R}_{r_k^{\top}} \delta u_k \right) + \frac{1}{2} \delta x_N^{\top} S \delta x_N + \underbrace{\bar{x}_N^{\top} S}_{q_N^{\top}} \delta x_N$$

Maximum Principle on Δ Cost

$$J = \frac{1}{2} \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + \frac{1}{2} x_N^T S_N x_N$$

$$x_{k+1} = A_k x_k + B_k u_k$$

Hamiltonian

$$\begin{aligned} H_k(x_k, u_k, \lambda_{k+1}) \\ = \frac{1}{2} x_k^T Q_k x_k + \frac{1}{2} u_k^T R_k u_k + \lambda_{k+1}^T (A_k x_k + B_k u_k) \end{aligned}$$

Maximum principle

$$\begin{aligned} \lambda_k &= \frac{\partial}{\partial x} H_k(x_k, u_k, \lambda_{k+1}) = Q_k x_k + A_k^T \lambda_{k+1} \\ 0 &= \frac{\partial}{\partial u} H_k(x_k, u_k, \lambda_{k+1}) = R_k u_k + B_k^T \lambda_{k+1} \end{aligned}$$

$$J_{\Delta} = \frac{1}{2} \sum_{k=0}^{N-1} (\delta x_k^{\top} Q \delta x_k + \delta u_k^{\top} R \delta u_k + 2 \underbrace{\bar{x}_k^{\top} Q}_{q_k^{\top}} \delta x_k + 2 \underbrace{\bar{u}_k^{\top} R}_{r_k^{\top}} \delta u_k) + \frac{1}{2} \delta x_N^{\top} S \delta x_N + \underbrace{\bar{x}_N^{\top} S}_{q_N^{\top}} \delta x_N$$

$$\delta x_{k+1} \approx A_k \delta x_k + B_k \delta u_k + d_k$$

Hamiltonian

$$\begin{aligned} H_k(\delta x_k, \delta u_k, \lambda_{k+1}) \\ = \frac{1}{2} \delta x_k^{\top} Q \delta x_k + \frac{1}{2} \delta u_k^{\top} R \delta u_k + \underbrace{q_k^{\top} \delta x_k + r_k^{\top} \delta u_k}_{\text{}} + \lambda_{k+1}^{\top} (A_k \delta x_k + B_k \delta u_k + d_k) \end{aligned}$$

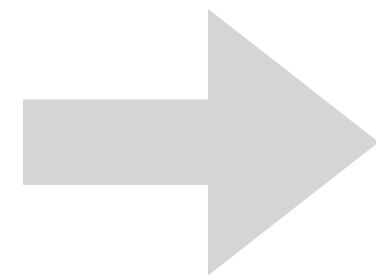
Maximum principle

$$\begin{aligned} \lambda_k &= \frac{\partial H_k}{\partial \delta x_k} = Q \delta x_k + \underbrace{q_k}_{\text{}} + A_k^{\top} \lambda_{k+1} \\ 0 &= \frac{\partial H_k}{\partial \delta u_k} = R \delta u_k + \underbrace{r_k}_{\text{}} + B_k^{\top} \lambda_{k+1} \end{aligned}$$

Backward Recursion

$$\lambda_k = \frac{\partial H_k}{\partial \delta x_k} = Q\delta x_k + \underline{q_k} + A_k^\top \lambda_{k+1}$$

$$0 = \frac{\partial H_k}{\partial \delta u_k} = R\delta u_k + r_k + B_k^\top \lambda_{k+1}$$



Due to the bias term in the backward recursion, the co-state becomes

$$\lambda_k = P_k \delta x_k + b_k$$

Where P_k comes from the Riccati equation

$$P_k = Q + A_k^\top P_{k+1} A_k - (A_k^\top P_{k+1} B_k)(R + B_k^\top P_{k+1} B_k)^{-1}(B_k^\top P_{k+1} A_k)$$

And the bias term becomes

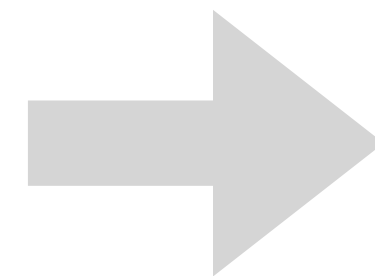
$$b_k = q_k + A_k^\top (P_{k+1} d_k + b_{k+1})$$

$$- (A_k^\top P_{k+1} B_k)(R + B_k^\top P_{k+1} B_k)^{-1} (r_k + B_k^\top (P_{k+1} d_k + b_{k+1}))$$

Optimal Δ Control Law

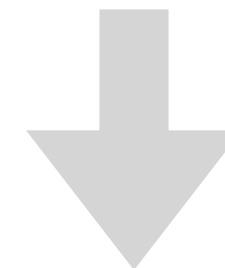
$$\lambda_k = \frac{\partial H_k}{\partial \delta x_k} = Q\delta x_k + q_k + A_k^\top \lambda_{k+1}$$

$$0 = \frac{\partial H_k}{\partial \delta u_k} = R\delta u_k + r_k + B_k^\top \lambda_{k+1}$$



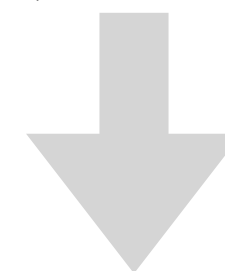
$$\lambda_{k+1} = P_{k+1}\delta x_{k+1} + b_{k+1}$$

$$= P_{k+1}[A_k\delta x_k + B_k\delta u_k + d_k] + b_{k+1}$$



$$0 = R\delta u_k + r_k + B_k^\top \lambda_{k+1} = R\delta u_k + r_k + B_k^\top P_{k+1}[A_k\delta x_k + B_k\delta u_k + d_k] + B_k^\top b_{k+1}$$

$$= [R + B_k^\top P_{k+1}B_k]\delta u_k + B_k^\top P_{k+1}A_k\delta x_k + B_k^\top P_{k+1}d_k + r_k + B_k^\top b_{k+1}$$



$$\delta u_k = -[R + B_k^\top P_{k+1}B_k]^{-1} B_k^\top P_{k+1}A_k\delta x_k - [R + B_k^\top P_{k+1}B_k]^{-1} [B_k^\top P_{k+1}d_k + r_k + B_k^\top b_{k+1}]$$

LQR Gain

**Additional term
due to errors in the
reference trajectory**

**Additional
term from
the
nominal
control**

**Additional
term from the
backward
recursion**

How to Mitigate Linearization Error?

$$\delta u_k = \underbrace{-[R + B_k^\top P_{k+1} B_k]^{-1} B_k^\top P_{k+1} A_k \delta x_k}_{\text{Feedback}} - \underbrace{[R + B_k^\top P_{k+1} B_k]^{-1} [B_k^\top P_{k+1} d_k + r_k + B_k^\top b_{k+1}]}_{\text{Feedforward}}$$

Feedback

Feedforward

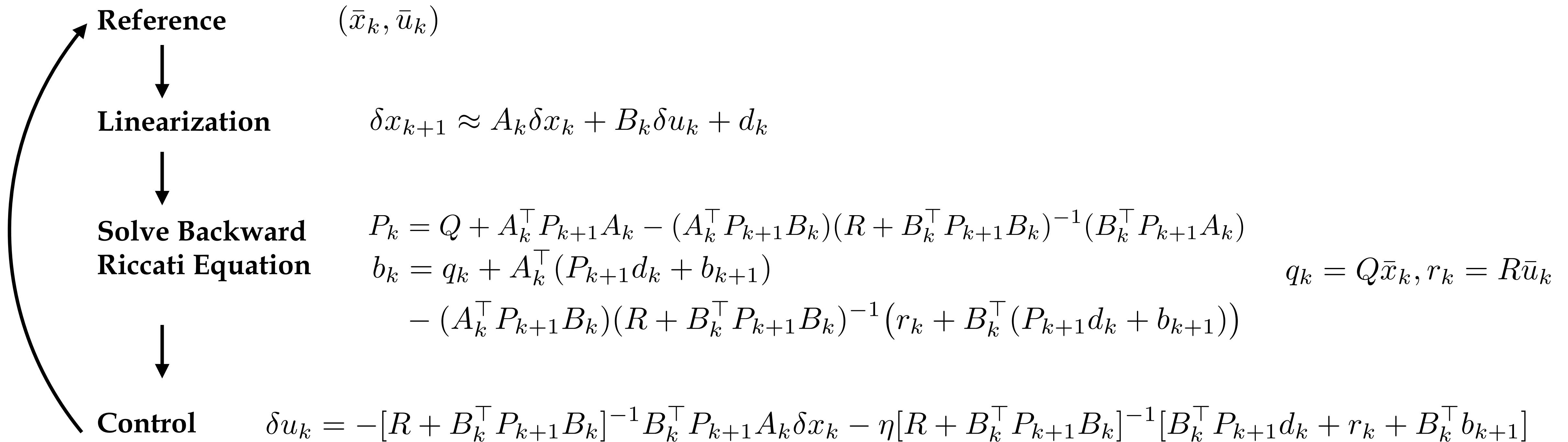


Usually need to scale this term to prevent instability due to linearization error (e.g., through line-search)



$$\delta u_k = -[R + B_k^\top P_{k+1} B_k]^{-1} B_k^\top P_{k+1} A_k \delta x_k - \eta [R + B_k^\top P_{k+1} B_k]^{-1} [B_k^\top P_{k+1} d_k + r_k + B_k^\top b_{k+1}]$$

Iterative Linear Quadratic Regulator



Remarks

- More generally, iLQR can take non-quadratic cost function. In that case, when computing the Δ cost, we will need to do quadratic approximation of the cost function as well.
- iLQR can be understood as a special case of Differential Dynamic Programming (DDP) with Gauss-Newton Hessian.